

# Opportunity & Objectives

The measurable targets this design is held to

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Objectives that cannot be measured cannot be failed, and objectives that cannot be failed are not objectives. Every target below is stated as a number, tied to the analysis that produces it, and reported against in this submission — including where the result is modest.

**Upload slot 01** — Opportunity & Objectives

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m<sup>2</sup> · Al Safa 2, Dubai

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Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

## 1. From ranked problems to measurable targets

The problem definition ranks thermal discomfort as the critical constraint. The objectives below follow from that ranking rather than from a general list of good intentions, and each one names the analysis that verifies it.

| Objective                                          | Target                  | Achieved             | Verified by                         |
|----------------------------------------------------|-------------------------|----------------------|-------------------------------------|
| Raise comfortable daylight hours                   | ≥ 60%                   | 64.6%                | 8,760-hour comfort model            |
| Provide a continuously shaded primary route        | ≥ 85% of daylight hours | 87.3%                | Ray-traced section across the year  |
| Minimise hours with no shade anywhere on the route | Minimise                | 52 h, from 330 h     | Plan-form sweep, src/config.py      |
| Reduce peak heat index                             | ≥ 5 °C                  | 7.13 °C mean         | Heat-index model, shaded vs exposed |
| Deliver within budget                              | ≤ AED 35 M              | AED 27.0 M (77%)     | Capital cost plan, src/costing.py   |
| Meet the brief's minimum programme                 | All items               | 20 facilities placed | Facilities map, src/plan.py         |

Each target is regenerated and re-checked on every pipeline run.

## 2. An objective the scheme deliberately does not maximise

**Site-wide mean shade is 34.1%, and no target was set for it.** Maximising average shade across 15,000 m<sup>2</sup> would spread the budget thin and produce a park that is uniformly mediocre. The scheme instead concentrates its shade into one continuous, genuinely excellent route. That is a design position, and it is stated here so the modest site-wide figure is read as intent rather than as failure.

## 3. Objectives beyond thermal comfort

- **Universal accessibility.** The site is level, the whole park is step-free, and the only change of level is ramped.
- **Day and night activation.** The convex face carries the evening uses; the crescent is lit from within.
- **Operational viability.** Commercial floor is placed to support running costs, and every facility is serviceable without a vehicle entering the walk.
- **Water discipline.** Open water is limited to a shaded channel; irrigation is subsurface drip on treated effluent.
- **Reproducibility.** Every claim in this submission can be regenerated from source data with one command.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.